



Synthesis, structure, Stokes and anti-Stokes visible luminescence of Tm^{3+} – Yb^{3+} co-doped $\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ nanorods

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Received: 3 August 2022 / Revised: 1 September 2022 / Accepted: 6 September 2022 / Published online: 4 October 2022
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Abstract

$\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ garnet nanorods co-doped with Tm^{3+} and Yb^{3+} optically active ions have been synthesized by Pechini sol-gel method, and their structural, morphological and luminescence properties have been explored. X-ray diffraction pattern confirmed that the synthesized $\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ nanomaterials were crystallized in cubic garnet structure with an average crystallite size of 30 nm. Morphology of these powders confirmed the formation of rods with an average length of 8 μm and diameter of 90 nm. The diffuse reflectance spectra were used to obtain partial energy-level structure of Tm^{3+} and Yb^{3+} ions in $\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ nanorods. Intense blue Stokes luminescence of Tm^{3+} ions at around 454 nm, under 360 nm ultraviolet excitation, as well as anti-Stokes luminescence at around 483 nm and co-operative luminescence at 490 nm, under near-infrared 980 nm excitation, have been measured. The intensity of these peaks increased with increasing of Yb^{3+} concentration up to 5 mol% and then decreased due to energy transfer between optically active ions. In addition, luminescence at ~545 and 565 nm has been observed in the anti-Stokes spectra which is due to the energy transfer from Ln^{3+} ions to the impurities such as Er^{3+} and Ho^{3+} ions. Laser pump power-dependent luminescence spectra confirmed that three photons were involved to populate the $^1\text{G}_4$ level in the upconversion mechanism. Luminescence decay curves have been measured for $^1\text{D}_2$ level and have been found to be non-exponential with an average lifetime of 20 μs . The non-exponential nature of decay curves is ascribed to energy transfer among Tm^{3+} ions through cross-relaxation processes. Colour co-ordinates of the samples excited at 980 nm lie in the blue–green region and also near to the white light region in the CIE diagram. All these results indicate that the present garnet nanorods could be a promising optically active material for the solid-state display devices.

Keywords $\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ garnet nanorods · Stokes and anti-Stokes luminescence · Upconversion · Tm^{3+} and Yb^{3+} ions

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1 Introduction

Upconversion is a non-linear optical process in which a photon of higher energy, visible or ultraviolet (UV), is emitted through the absorption of two, or more, lower energy near-infrared (NIR) photons. Upconverting nanoparticles have a wide range of applications in solar energy conversion, photocatalysis, three-dimensional display, bio-imaging, optical temperature and pressure sensors, etc. [1–5]. The upconversion process involves ladder like energy-level structure that can be realized with lanthanide (Ln^{3+}) ions. Ln^{3+} -doped nano-crystalline materials exhibit its potentiality in bio-imaging applications owing to their remarkable properties including high photo stability, low bio-toxicity, large anti-Stokes shifts, narrow emission bandwidths and low background auto-fluorescence [6, 7]. It is well known that Tm^{3+} , Ho^{3+} and Er^{3+} ions have suitable ladder type

energy-level structure to promote upconversion and act as very good upconverters.

Among Ln^{3+} ions, Tm^{3+} ion is one of the efficient activators that can be excited by both UV and NIR radiation, and the Stokes and anti-Stokes luminescence spectra of Tm^{3+} -doped materials generally contain also multi-band emissions ranging from UV to NIR [8, 9]. Of special interest is the $\text{Tm}^{3+} {}^1\text{D}_2 \rightarrow {}^3\text{F}_4$ transition, at around 450 nm, due to its high capacity for data storage optical devices. To enhance the upconversion efficiency of Tm^{3+} ions in a given host matrix, Yb^{3+} ion is the most considerable sensitizer that provides a broad absorption cross-section in the NIR region. Very recently, Ge et al. [10] reported the blue upconversion luminescence from $\text{Bi}_2\text{Ti}_2\text{O}_7:\text{Tm}^{3+}/\text{Yb}^{3+}$ nanofibers and its potential for “in vivo” temperature sensing applications in biological tissues. In addition, Dwivedi et al. [11] prepared $\text{Tm}^{3+}/\text{Yb}^{3+}$ co-doped upconverted phosphors and observed laser-induced optical heating that depends directly on the pump power.

Upconverted luminescence properties, such as luminescence intensity and peak profiles, directly depend on the crystal-field interaction felt by the Ln^{3+} ions, multiphoton and energy transfer processes in a host material. Therefore, selection of a host material with low phonon energy is essential to obtain high luminescence efficiency. The low phonon energy fluorides, of toxic nature, are regarded as the most accepted upconverted host material [12]. However, for some specific applications, oxides are also most promising non-toxic upconverting materials because of their better physical, thermal and chemical stabilities [13]. Among diverse oxide-based host materials, garnets of general formula $\text{A}_3\text{B}_5\text{O}_{12}$ are found to be one of the best host matrices for the development of laser, photonic and light emitting diodes [14]. They possess superior optical, chemical, thermal and mechanical properties such as high density, high thermal conductivity, hardness and structural stability [15]. It is well known that, among different families of garnets, lutetium-based garnets containing gallium, such as $\text{La}_3\text{Lu}_2\text{Ga}_3\text{O}_{12}$, $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ or $\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ are particularly promising materials for photonic device applications, as they have greater Ln^{3+} ion solubility because of their higher unit cell volume, low concentration quenching, high refractive index, larger stimulated emission cross-sections and high density [16]. Moreover, the luminescence intensity of Ln^{3+} ions in Lu-based garnets is expected to increase due to the intensity-borrowing mechanism [17]. It is worth to mention that research on Er^{3+} -doped garnets proved that the $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ host has superior upconversion properties compared to well-known $\text{Y}_3\text{Al}_5\text{O}_{12}$ host matrix [18].

A disordered mixed-garnet structure results in inhomogeneous broadening consequently, the absorption cross-section and the emission spectral range may be enhanced. Stokes and anti-Stokes photoluminescence in $\text{Lu}_3\text{Ga}_5\text{O}_{12}$

nano-garnets co-doped with $\text{Tm}^{3+}/\text{Yb}^{3+}$ ions have already been reported [19] and, in this work, the composition of the matrix have been modified by replacing part of trivalent gallium (Ga^{3+}) in octahedral sites by trivalent scandium (Sc^{3+}). Thus, a distortion of the garnet lattice due to different ionic radii of Sc^{3+} and Ga^{3+} ions is expected, which may further enhance the emission intensity from the Ln^{3+} ions emitting levels via excitation energy transfer to Ln^{3+} ions, thanks to the good spectral overlap of both Sc^{3+} -emission and Ln^{3+} -absorption spectral bands [20]. For example, Sato et al. [21] observed that the emission bandwidths of $\text{Y}_3\text{ScAl}_4\text{O}_{12}:\text{Nd}^{3+}$ ceramics were enhanced five times than that of $\text{Y}_3\text{Al}_5\text{O}_{12}:\text{Nd}^{3+}$. Furthermore, Ln^{3+} ions experiences tighter lattice in $\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ (LuSGG) (12.26 Å), since its lattice constant is slightly smaller than that of $\text{Y}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ (12.42 Å), giving rise to stronger crystal-field interactions between the optically active ions and its ligands [22], which improved the population inversion and decreased the lasing threshold at room temperature [23]. Wu et al. [24] reported, for the first time, the passively Q-switched LuSGG: Nd^{3+} crystal with the shortest laser pulse width, largest pulse energy and highest peak power. In addition, LuSGG: Nd^{3+} crystal is also identified as a good promising laser material for small emission cross-sections, excellent pulsed laser performance, and also broad emission spectrum [24, 25].

Recently, our group has reported the co-operative upconverted luminescence in LuSGG: Yb^{3+} nano-crystalline powders and it has been observed that large ground-state splitting in this host favours the generation of ultra-short pulses and high power femtosecond laser at 1 μm wavelength [26]. In this paper, the authors are interested in studying the visible Stokes and anti-Stokes luminescence of $\text{Tm}^{3+}-\text{Yb}^{3+}$ co-doped LuSGG garnet nanorods for solid-state lighting applications. So far, very few reports are available for the Ln^{3+} -doped LuSGG nanomaterials and, to the best of authors' knowledge, no studies/reports are available on the present study. Therefore, the suitability of the present $\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}:\text{Tm}^{3+}-\text{Yb}^{3+}$ nano-garnet powders for NIR excited pc-LEDs that can transform NIR to the visible and UV light has been investigated.

2 Experimental and characterization techniques

Tm^{3+} and Yb^{3+} co-doped lutetium scandium gallium, $\text{Lu}_{3-0.03-x}\text{Sc}_2\text{Ga}_3\text{O}_{12}:0.03\text{Tm}^{3+}-x\text{Yb}^{3+}$ (LuSGG: $\text{Tm}-\text{Yb}$) garnet nanorods, where $x = 0, 0.03, 0.15$ and 0.30 (representing Tm^{3+} : 1 and Yb^{3+} : 0, 1, 5 and 10 mol%), were synthesized by the sol-gel method [27]. For convenience, these nanorods are labelled hereafter as LuSGG:1Tm, LuSGG:1Tm-1Yb, LuSGG:1Tm-5Yb and LuSGG:1Tm-10Yb. All chemicals were taken from Sigma-Aldrich with

high purity ($\geq 99.9\%$). The raw materials of lutetium (III) nitrate hexahydrate, thulium (III) nitrate pentahydrate, ytterbium (III) nitrate pentahydrate, scandium (III) nitrate hydrate, gallium (III) nitrate hydrate were dissolved in the 25 ml of 1 M HNO_3 while stirring. Detailed explanations of the experimental procedure, as well as the analytical and optical techniques for the present study were given in our earlier paper [26]. X-ray diffraction (XRD) patterns were recorded using a diffractometer (RIGAKU, Miniflex-600) using $\text{Cu K}\alpha$ radiation (1.5406 Å) in the range of $2\theta = 15^\circ$ – 60° , with a step size of 0.02° . CARL ZEISS EVO MA15 SEM was used to obtain scanning electron microscope (SEM) images and energy-dispersive spectra (EDS) measurements. The diffuse reflectance in UV-VIS-NIR region was measured with a spectrophotometer (Agilent Technologies Cary 5000). Edinburgh UV-Vis-NIR (FLS 980) spectrometer with xenon arc lamp (250 W) and 980 nm laser diode as excitation sources was used for photoluminescence spectral measurements of $\text{LuSGG}:\text{Tm}^{3+}\text{-Yb}^{3+}$ garnet nanorods.

3 Results and discussion

3.1 Structure and morphology

XRD patterns of synthesized nanorods have been measured, exhibit similar patterns, and are displayed in Fig. 1 for LuSGG garnets doped with 1.0 mol% of Tm^{3+} and co-doped with 1.0 mol% of Tm^{3+} and Yb^{3+} ions. Patterns contain sharp peaks, indicating well-crystallized phases in a cubic garnet structure and good agreement with $la\text{-}3d$ (No. 230) space group, as per JCPDS no. #54-1252 (or ICDD # 04-001-7112) [28, 29]. In addition, a few unknown by-product peaks were noticed at

Bragg angles $\sim 21.5^\circ$, 30.6° , 37.9° and 42° due to the excess amount of Sc_2O_3 , matched with the JCPDS no. #74-1210 [30], as observed in $\text{LuSGG}:\text{Yb}^{3+}$ [26] and $\text{Y}_3\text{Sc}_2\text{AlGa}_2\text{O}_{12}:\text{Nd}^{3+}$ nano-powders [31]. Though the sample contains Sc_2O_3 secondary phase and its content is small in the LuSGG host based on intensity of XRD peaks. The incorporation of Tm^{3+} and Yb^{3+} ions into $\text{Lu}_3\text{Sc}_2\text{Ga}_3\text{O}_{12}$ or Sc_2O_3 depends on the size and electronegativity of atoms or ions. The radii (0.885 Å) and electronegativity (1.36) of Sc^{3+} ion are largely different from those of Lu^{3+} (1.001 Å, 1.27), Tm^{3+} (1.02 Å, 1.25) and Yb^{3+} (1.008 Å, 1.1) ions. Hence, it is presumed that the $\text{Tm}^{3+}/\text{Yb}^{3+}$ ions occupy the dodecahedral Lu^{3+} site instead of Sc^{3+} site in the present host. In the case of $\text{Cr}^{3+}:\text{LuSGG}$, it was predicted that Cr^{3+} ion prefers to occupy Sc^{3+} site rather than Lu^{3+} site due to similar radii and electronegative values [29].

The average crystallite size (D) is calculated using Debye–Scherrer formula

$$D = \frac{k\lambda}{\beta \cos \theta} \quad (1)$$

where λ is the wavelength of X-ray; β is the full width at half maximum (FWHM); θ is the angle of diffraction peak; and k is the shape factor (0.89). The crystallite size, interplanar spacing and lattice constant for $\text{LuSGG}:\text{1Tm}$ sample are found to be 30 nm, 2.754 Å and 12.316 Å, whereas for $\text{LuSGG}:\text{1Tm-1Yb}$ sample are only slightly different, i.e., 29 nm, 2.755 Å and 12.321 Å, respectively. The crystallite size of $\text{LuSGG}:\text{1Tm-1Yb}$ sample is found to be similar to that of $\text{LuGG}:\text{1Tm-1Yb}$ [19] and $\text{LuSGG}:\text{Yb}$ [26], but it is noticed that the lattice parameter is slightly higher than that of $\text{LuGG}:\text{Tm-Yb}$ (12.241 Å) and lower than that of $\text{LuSGG}:\text{Yb}$ (12.383 Å) [26]. This is due to addition of Sc into the host and variations in the positions of oxygen ions which can be easily polarizable.

From SEM images of $\text{LuSGG}:\text{1Tm}$ and $\text{LuSGG}:\text{1Tm-1Yb}$ samples, shown in Fig. 2a–c, it is concluded that the sample consists of nanorods of length varying from ~ 5 to 8 μm and diameter from ~ 90 to 150 nm. The similar type of morphology was also reported in the Tm^{3+} and Yb^{3+} co-doped zirconium titanate nano-composites synthesized by solution combustion method [32]. The EDS gives the information about the chemical composition of the given nanorods, shown in Fig. 3a, b for $\text{LuSGG}:\text{1Tm}$ and $\text{LuSGG}:\text{1Tm-1Yb}$ nanorods along with their weight and atomic percentages, in which it is observed that six elements, Lu^{3+} , Sc^{3+} , Ga^{3+} , O^{2-} , Tm^{3+} and Yb^{3+} , are distributed uniformly in the $\text{LuSGG}:\text{Tm-Yb}$ nanorods and no other impurities have been found in these samples. It is also noticed that Sc at. wt% is slightly high in $\text{LuSGG}:\text{1Tm-1Yb}$ sample, in accordance with the XRD results.

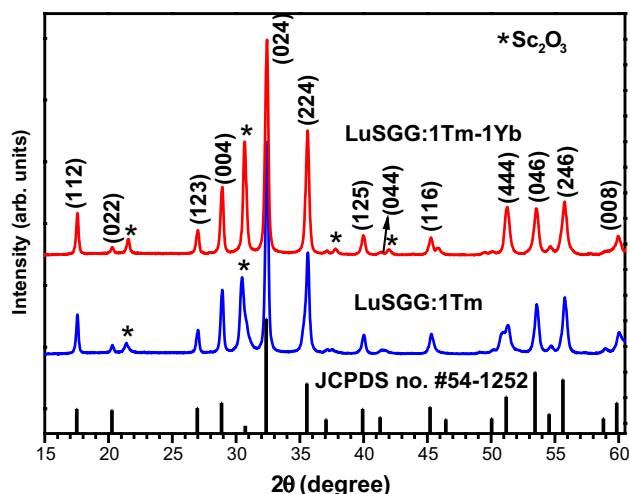


Fig. 1 XRD patterns of $\text{LuSGG}:\text{1Tm}$ and $\text{LuSGG}:\text{1Tm-1Yb}$ along with the JCPDS data

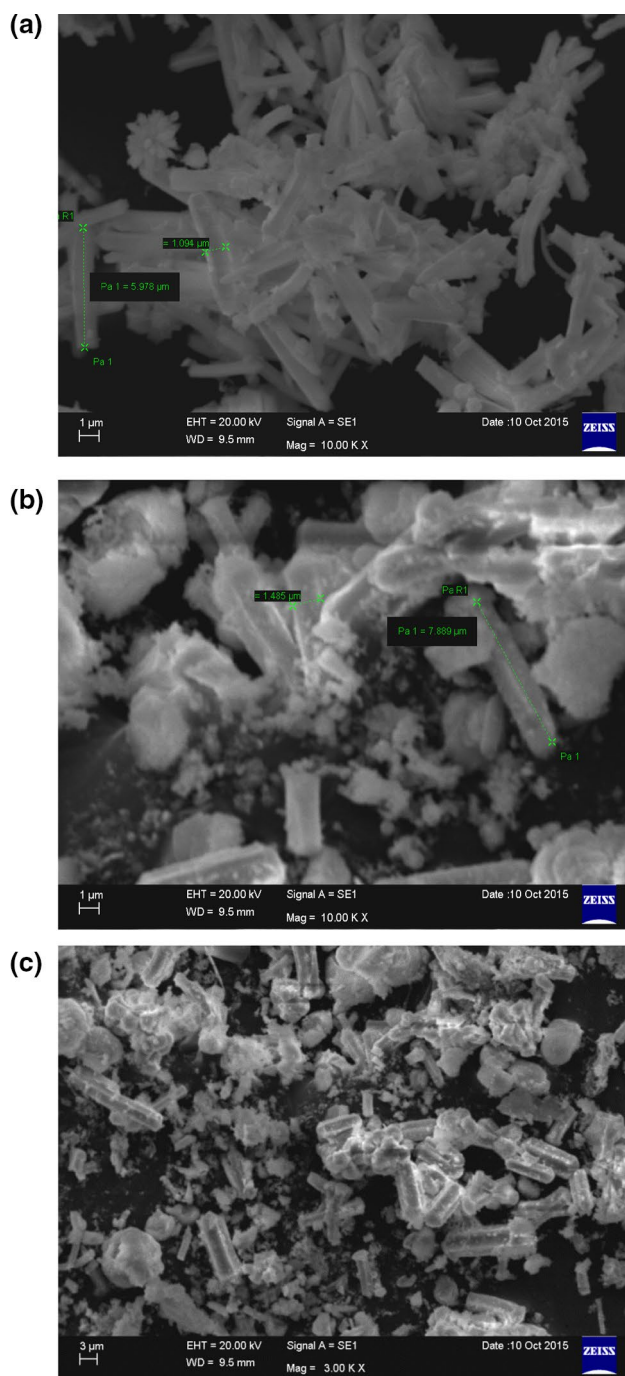


Fig. 2 SEM images of **a** LuSGG:1Tm and **b, c** LuSGG:1Tm-1Yb nanorods

3.2 Diffuse reflectance spectra

The diffuse reflectance spectra of LuSGG:1Tm and LuSGG:1Tm-10Yb garnet nanorods in the region of (a) 300–800 nm and (b) 880–2000 nm are shown in Fig. 4. Inset of Fig. 4a shows the enlarged spectrum in the 460–500 nm range. The spectrum is similar to that

of Tm³⁺/Yb³⁺-doped LuGG [19], BaLa₂ZnO₅ [33], and YPO₄ [34]. The diffuse reflectance spectrum of LuSGG:1Tm garnet nanorods consists of 20 absorption peaks assigned to ³F₄ → ¹I₆ + ³P_J (335, 346 nm), ³H₆ → ¹D₂ (362 nm), ³H₅ → ¹I₆ + ³P_J (387, 419 nm), ³F₄ → ¹D₂ (449, 457, 465 nm), ³H₆ → ¹G₄ (474, 486, 491, 495 nm), ³H₅ → ¹D₂ (538 nm), ³F₄ → ¹G₄ (637, 650 nm), ³H₆ → ³F₂ (656, 665 nm), ³H₆ → ³H₅ (1133, 1204 nm) and ³H₆ → ³F₄ (1926 nm) transitions of Tm³⁺ ion [19, 35–37].

When the Yb³⁺ ions are introduced into the LuSGG:Tm³⁺, a broad absorption band is observed in the region 890–1050 nm (see Fig. 4b) attributed to the ²F_{7/2} → ²F_{5/2} transition of Yb³⁺ ion. The main peak is located at around 975 nm with a bandwidth of around 9.65 nm. This band is similar to our earlier report on LuSGG:Yb [26], but the bandwidth of the 975 nm peak is observed to be slightly higher in LuSGG:1Tm-10Yb compared to LuSGG:20Yb (9.05 nm) [26]. The peaks observed are attributed to the transitions between Stark splitting of Yb³⁺ ion and, as given in Ref. [26], ground state ²F_{7/2} splits into four multiplets (1, 2, 3, and 4), whereas excited state into three multiplets (5, 6 and 7). Thus, peaks at 898, 928, 975, 996 and 1026 nm correspond to the transitions between 1 → 7, 1 → 6, 1 → 5, 3 → 5 and 4 → 5 of the ²F_{7/2} ground state and the ²F_{5/2} excited multiplet, respectively. It is worth noting that no absorption band is seen at around 980 nm in the single-doped LuSGG:1Tm, which demonstrate that only Yb³⁺ ions can be excited with 980 nm laser excitation. Moreover, the peaks located in the 900–980 nm region show well matching with the emission radiation band of commercial, low-cost InGaAs diode laser, indicating the possibility of using this nanomaterial for solid-state lighting applications.

Besides these Yb³⁺ peaks, a few more peaks at 684 and 790 nm were observed in the spectrum of LuSGG:1Tm-10Yb associated with the ³H₆ → ³F₃ and ³H₆ → ³H₄ transitions of Tm³⁺ ion, respectively. Rathiah et al. [19] also observed the same ³H₆ → ¹D₂, ¹G₄, ³F₂, ³F₃, ³H₄, ³H₅ and ³F₄ transitions when a concentration of 10 mol% of Yb³⁺ ions were introduced into the LuGG:1Tm host, with relatively decreased intensity. Xie et al. [33] observed only ³H₆ → ³F₂, ³F₃ and ³H₄ transitions with suppressed intensity when 10 mol% of Yb³⁺ ions was added to the BaLa₂ZnO₅:0.75Tm host. Zheng et al. [34] observed the ³H₆ → ¹G₄, ³F₂, ³F₃, ³H₄, ³H₅ and ³F₄ transitions when 10 mol% of Yb³⁺ ions was introduced into the YPO₄:1Tm host. The suppression of Tm³⁺ peaks in the diffuse reflectance spectrum of the co-doped matrix is due to high absorption cross-section of Yb³⁺ ions. However, Stokes emission in the NIR region with different excitations would give more insights about the energy transfer between Tm³⁺ and Yb³⁺ ions, which is beyond the scope of the present paper. Furthermore, the broad band in the UV region (overlapped with 362 nm) may be due to defect O²⁻–Yb³⁺ charge transfer band [38]. These spectra

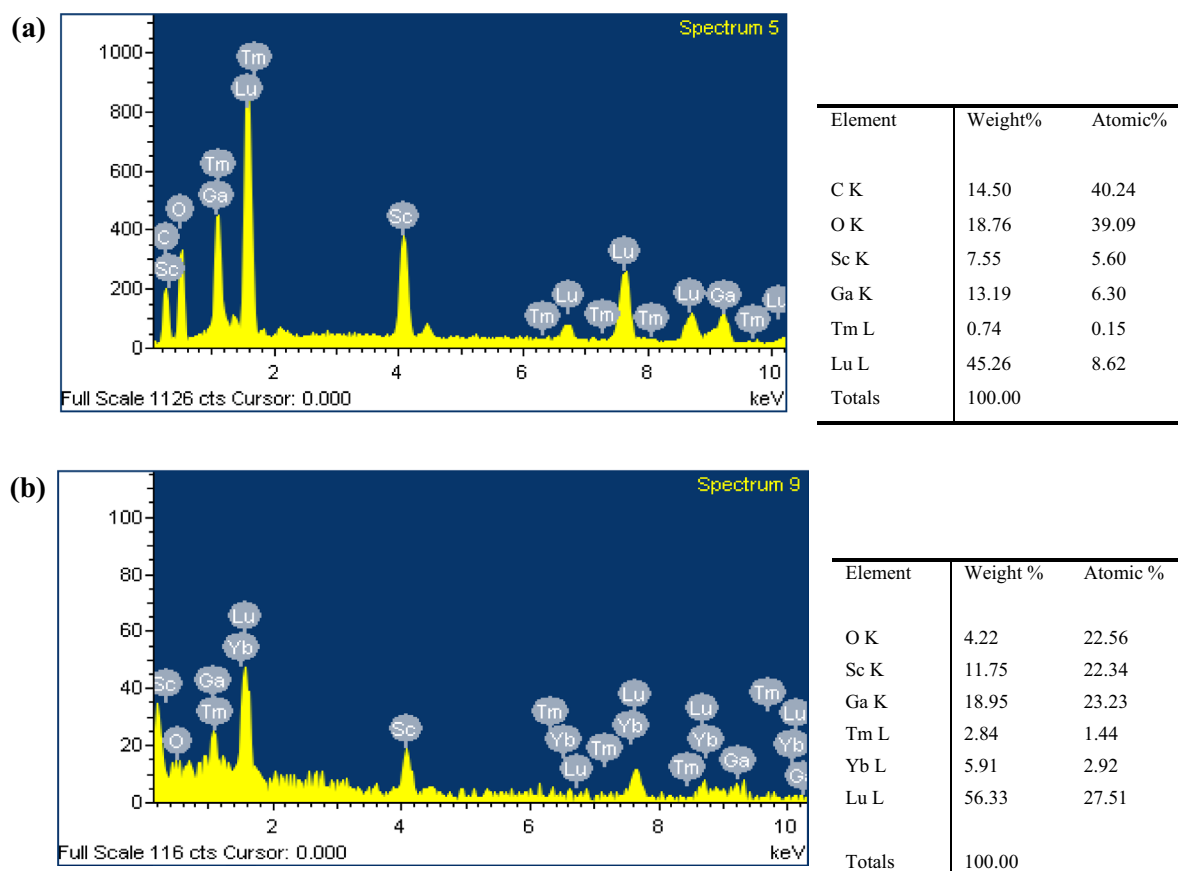


Fig. 3 EDS of **a** LuSGG:1Tm and **b** LuSGG:1Tm-1Yb nanorods along with weight % and atomic % of their presence in the nanorods

have been used to derive the partial energy-level diagram of Tm³⁺, Yb³⁺-doped LuSGG garnet nanorods and is shown in the subsequent sections.

3.3 Photoluminescence studies

3.3.1 Stokes emission

Excitation and Stokes luminescence spectra of LuSGG:1Tm-xYb nanorods are shown in Fig. 5a. Excitation spectra, recorded by monitoring the emission at 454 nm, consist of three bands due to the excitation of Tm³⁺ ions from the ³H₆ ground state to the ¹I₆ + ³P_J (304 and 315 nm) and ¹D₂ (360 nm) excited multiplets, respectively. When the nanorods are excited at 360 nm, the luminescence spectrum consists of a broad band extending from 400 to 530 nm, in which sharp peaks associated with the ¹D₂ → ³F₄ (454, 458, 462, 466 nm), ¹G₄ → ³H₆ (473 and 483 nm), and ¹D₂ → ³H₅ (514 nm) transitions of Tm³⁺ ion [32, 39]. The sharpness of the peaks indicates the good crystalline nature of the samples as well the occupation of Ln³⁺ ions at the Lu³⁺ sites. Inset of Fig. 5a shows the extended portion of the luminescence spectrum of LuSGG:1Tm-5Yb in the region 400–750 nm.

These luminescent peaks between 454 and 483 nm are associated with the Stark splitting of ¹D₂ and ¹G₄ multiplets [39], suggesting that the Tm³⁺ ions experiences a relatively strong crystal-field strength in the LuSGG:Tm³⁺-Yb³⁺ nanorods under study compared to other hosts [35, 39]. It seems that the degeneracy of ¹D₂ level is completely lifted, as observed in the LuGG:Tm-Yb host matrices [19], and it is unusual in other non-garnet hosts.

In addition to these, very weak peak at 491 nm is also observed in the Stokes spectra, and corresponds to the co-operative luminescence from the Yb³⁺ ions exciting at 360 nm. A stronger Coulombic interaction between two closely spaced Yb³⁺ ions leads to co-operative luminescence in the visible region. It is a special upconversion process in which two excited Yb³⁺ ions simultaneously return to the ground state giving rise to one higher energy photon due to adjacent ion pairs by multi-polar interaction, magnetic-dipole interaction or exchange coupling [40]. The presence of this peak in the Stokes spectra with 360 nm excitation indicates that very small amount of energy is being transferred from the Tm³⁺ ion to the Yb³⁺ one. One possibility to observe this peak is that when the ³H₄ and ³H₅ levels are populated through non-radiative relaxations of the excited

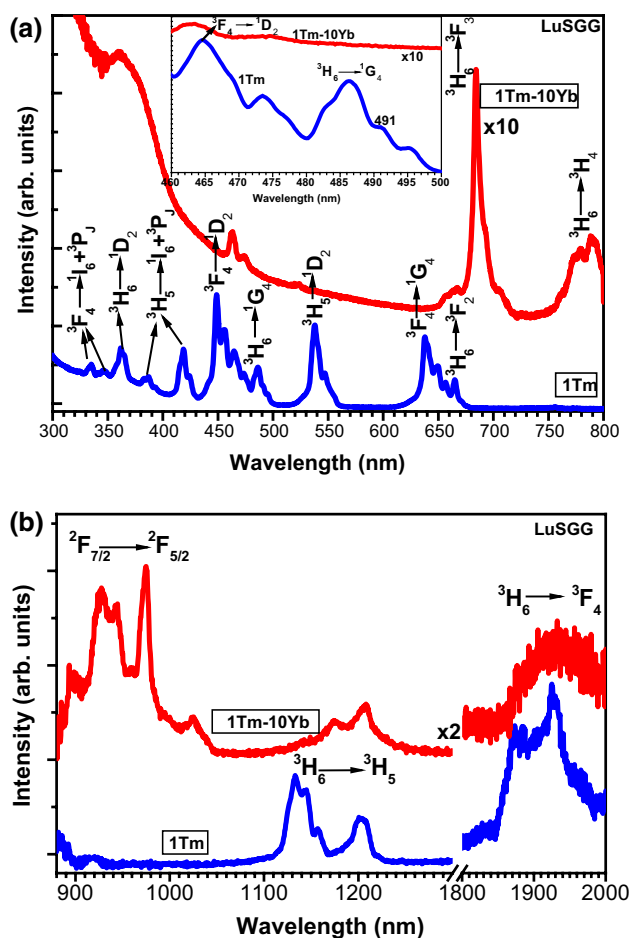


Fig. 4 Diffuse reflectance spectra of LuSGG:1Tm and 1Tm-10Yb nanorods in **a** 300–800 nm and **b** 890–2000 nm regions. Inset in **a** shows the enlarged spectra in the region 460–500 nm

level, the energy transfer to the $^2F_{5/2}$ state of Yb^{3+} ions can take place. This non-resonant energy transfer between Tm^{3+} and Yb^{3+} ions occurs through phonon-assisted energy transfer, as the 3H_4 and 3H_5 levels are slightly above and below the average energy of the $^2F_{5/2}$ level [41]. The excited Yb^{3+} ions might be coupled together and gives rise to co-operative luminescence. Another possibility might be that part of visible energy at 473/483 nm of Tm^{3+} ion is transferred to the two Yb^{3+} ions (980 nm) and excite them to $^2F_{5/2}$ level, leading to Yb–Yb coupling and giving raise to the co-operative luminescence at 491 nm. However, the luminescence spectra in the NIR region may give more clarity on this energy transfer. However, this energy transfer is not efficient in the LuSGG:Tm–Yb samples, as the intensity of 491 nm peak is very low.

On the other hand, this 491 nm peak is distinctly visible even in the single Tm^{3+} -doped LuSGG garnet nanorods. This peak is attributed to the transition from one of the multiplets of 1G_4 to the ground state, 3H_6 (i.e., $^1G_4 \rightarrow ^3H_6$). This

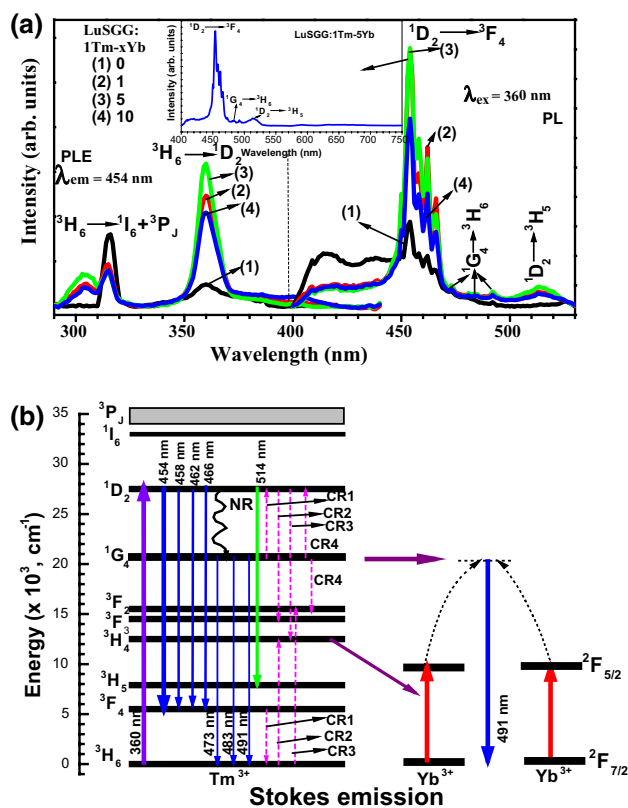


Fig. 5 **a** Excitation and PL spectra of LuSGG:1Tm- x Yb nanorods. Inset shows the expanded emission spectrum of LuSGG:1Tm-5Yb and **b** partial energy-level diagram showing the Stokes emission, non-radiative (NR) and possible cross-relaxation channels (CR)

can also be confirmed from the diffuse reflectance spectra in the inset of Fig. 4a. There is a peak at 491 nm in the spectrum of Tm^{3+} -doped LuSGG, which is absent in that of Tm^{3+} – Yb^{3+} -doped LuSGG sample. Hence, it is concluded that the peak at 491 nm in the luminescence spectrum of single-doped sample is due to the $^1G_4 \rightarrow ^3H_6$ transition of Tm^{3+} ion, and in the co-doped samples, it could be due to both the $^1G_4 \rightarrow ^3H_6$ transition of Tm^{3+} ion and co-operative luminescence from the Yb^{3+} ions. Moreover, the intensity of this peak is varied with the Yb^{3+} ion concentration in the co-doped samples similar to that of the other peaks.

Furthermore, the peak intensity of excitation and luminescence spectra is enhanced with the increase of the Yb^{3+} concentration up to 5 mol%, and then reduced for 10 mol%. Therefore, the optimum concentration is 5 mol% of sensitizer (Yb^{3+}) ion in LuSGG:1Tm- x Yb samples. The enhancement in the luminescence intensity with increasing sensitizer (Yb^{3+}) concentration could be due to the resonant electronic transitions $^1G_4 \leftrightarrow ^3H_4$, $^3F_2 \leftrightarrow ^3F_4$, $^3H_5 \leftrightarrow ^3H_6$ and $^1D_2 \leftrightarrow ^3P_2$ of Tm^{3+} ion with $^2F_{5/2} \leftrightarrow ^2F_{7/2}$ of Yb^{3+} ion, followed by Tm^{3+} – Tm^{3+} cross-relaxations (CR) [39]. The possible cross-relaxation channels in the Tm^{3+} ion are: **CR1**: (1D_2 ,

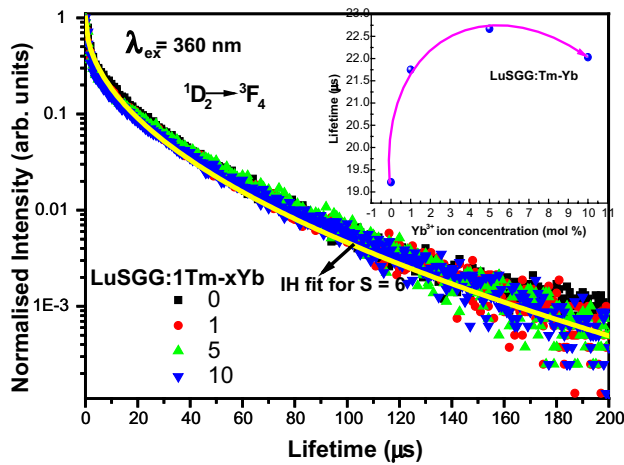


Fig. 6 Decay profiles for $^1\text{D}_2 \rightarrow ^3\text{F}_4$ transition of Tm^{3+} ion in LuSGG:Tm-Yb nanorods. IH fit for $S=6$ for LuSGG:1Tm-10Yb is included. Inset shows the variation of lifetime ($^1\text{D}_2$ level) with respect to Yb^{3+} ion concentration and solid line is to guide the view

$^3\text{H}_6 \rightarrow ^1\text{G}_4$, $^3\text{F}_4$), **CR2**: ($^1\text{D}_2$, $^3\text{H}_6 \rightarrow ^3\text{F}_3$, $^3\text{H}_4$), **CR3**: ($^1\text{D}_2$, $^3\text{H}_6 \rightarrow ^3\text{H}_4$, $^3\text{F}_2$) and **CR4**: ($^1\text{G}_4$, $^1\text{G}_4 \rightarrow ^3\text{F}_2$, $^1\text{D}_2$), as shown in Fig. 5b.

The energy-level diagram of LuSGG:Tm-Yb garnet nanorods explaining the Stokes emission is depicted in Fig. 5b. In this case, with 360 nm excitation, the Tm^{3+} ions are initially excited resonantly to the $^1\text{D}_2$ excited state from the $^3\text{H}_6$ ground state, which populates the $^1\text{D}_2$ level of Tm^{3+} ion. From this multiplet, the radiative transitions $^1\text{D}_2 \rightarrow ^3\text{F}_4$ (454, 458, 462, 466 nm) and $^1\text{D}_2 \rightarrow ^3\text{H}_5$ (514 nm) emissions take place. In addition to these radiative transitions, some of the Tm^{3+} ions get relaxed to $^1\text{G}_4$ level non-radiatively (NR) and gives raise to the radiative transitions $^1\text{G}_4 \rightarrow ^3\text{H}_6$ (473, 483, 491 nm) from the multiplets of $^1\text{G}_4$. Inefficient energy transfer can be taken place through $\text{Tm}^{3+} \rightarrow \text{Yb}^{3+}$ that results weak co-operative luminescence at 491 nm.

The fluorescence decay curves for the $^1\text{D}_2$ level of Tm^{3+} ion in LuSGG:1Tm-xYb nanorods under 360 nm excitation wavelength and monitoring the $^1\text{D}_2 \rightarrow ^3\text{F}_4$ transition at 454 nm are shown in Fig. 6.

It is clear that all the decay curves exhibit non-exponential nature and, hence, the average lifetime has been evaluated by the following equation:

$$\tau_{\text{avg}} = \frac{\int I(t) dt}{\int t dt} \quad (2)$$

The lifetime of $^1\text{D}_2$ level of Tm^{3+} ion calculated using above equation is found to be slightly increased from 19 to 23 μs with escalating Yb^{3+} ion concentration up to 5 mol% and then quenches, as represented in the inset of Fig. 6. The emission decay times are found to be slightly lesser than that of $\text{Tm}^{3+}/\text{Yb}^{3+}$ -doped LaAlO_3 nano-crystals (30 μs) [35]. The

non-exponential nature of decay profiles proved the existence of energy transfer due to concentration quenching that is taking place between donor and acceptor ions. To assess the energy transfer processes, the decay profiles of $^1\text{D}_2$ level were studied in the frame work of Inokuti-Hirayama (IH) model [42]

$$I(t) = I_0 \exp \left\{ -\frac{t}{\tau_0} - Q \left(\frac{t}{\tau_0} \right)^{3/S} \right\} \quad (3)$$

where ' t ' is the time after excitation, ' τ_0 ' is the intrinsic decay time of the donors in the absence of acceptors and ' Q ' is the energy transfer parameter. The value of S ($=6, 8$ or 10) depends on whether the dominant mechanism of the interaction is dipole-dipole, dipole-quadrupole or quadrupole-quadrupole, respectively. The decay profiles are well fitted to Eq. (3) for $S=6$, as shown in Fig. 6, indicating that the energy transfer between donor and acceptor is of dipole-dipole type. From the fit, energy transfer parameter ' Q ' is found to be around 0.24 for LuSGG:Tm-Yb samples. This energy transfer could be due to the cross-relaxation between Tm^{3+} ions as well as $\text{Tm}^{3+}\text{-Yb}^{3+}$ interactions.

3.3.2 Anti-Stokes emission

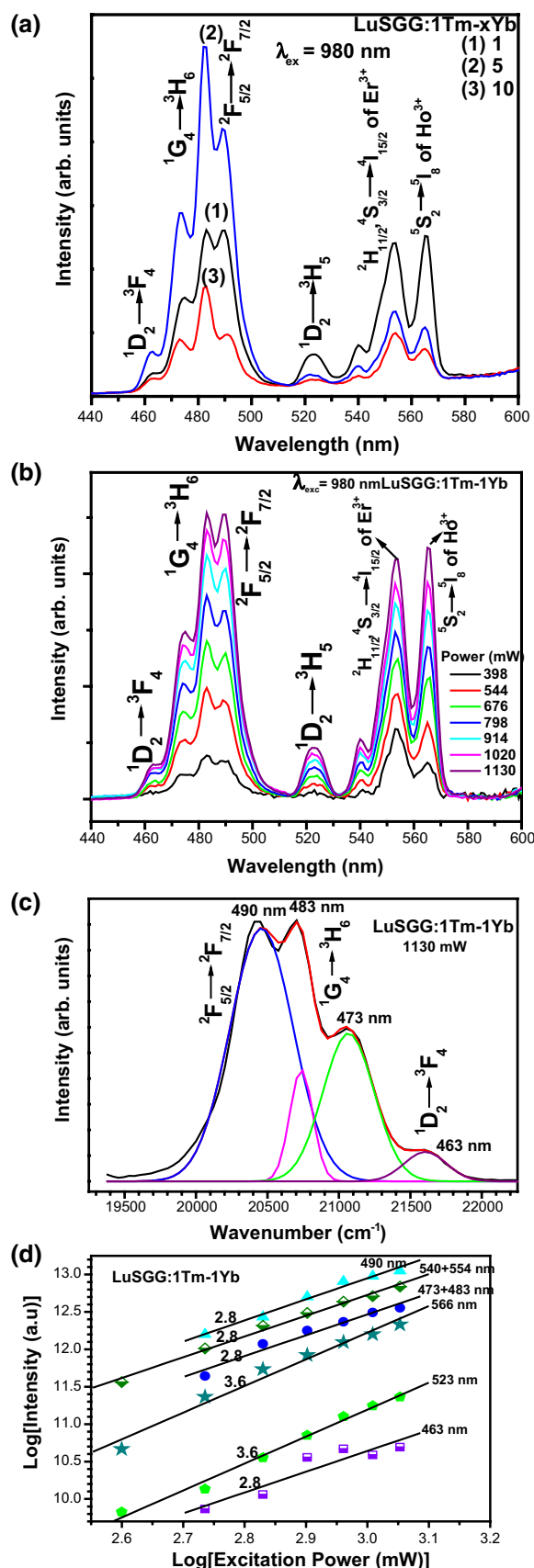
Figures 7 depicts the anti-Stokes spectra of LuSGG:Tm-Yb nanorods in the wavelength region of 440–600 nm under 980 nm laser excitation. Figure 7a shows the concentration-dependent upconversion spectra, whereas Fig. 7b presents the power-dependent upconversion spectra of LuSGG:1Tm-1Yb nanorods. Both the spectra consist of several peaks at $\sim 463, 473, 483, 490, 523, 540, 554$ and 566 nm spread in blue and green spectral region. Figure 7c shows the de-convolution of peaks at 463, 473, 483 and 490 nm. In the anti-Stokes spectra, shown in Fig. 7a, b, the peaks at 463 nm correspond to $^1\text{D}_2 \rightarrow ^3\text{F}_4$ transition, 473 and 483 nm correspond to $^1\text{G}_4 \rightarrow ^3\text{H}_6$ transition (Stark splitting of $^1\text{G}_4$ level) and 523 nm arises due to $^1\text{D}_2 \rightarrow ^3\text{H}_5$ transition of Tm^{3+} ion. These radiative transitions are originated by the energy transfer upconversion (ETU) from sensitizer (Yb^{3+})-to-activator (Tm^{3+}) ion. Comparing the Stokes and anti-Stokes spectra, the relative peak positions are found to be same in the blue region, except in their relative intensities. Unlike in the Stokes spectra, it is noticed in the anti-Stokes spectra that very intense peak is at 483 nm, whereas weak peak is at 463 nm (relative peak intensities are different). Also, the strong peak at 490 nm is attributed to co-operative luminescence from Yb^{3+} ion. Under pumping of 980 nm laser diode, Yb^{3+} ions are moved to the $^2\text{F}_{5/2}$ energy level, from which two of these excited Yb^{3+} ions are coupled together to attain a effective state in the region of 490 nm through co-operative upconversion energy transfer process.

Fig. 7 Upconversion spectra of **a** LuSGG:Tm–Yb for different concentrations of Yb^{3+} ions at 1130 mW, **b** LuSGG:1Tm–1Yb for different pump powers, **c** de-convoluted blue region of the spectra given in **b** and **d** excitation power density dependency of blue and green intensities of LuSGG:1Tm–1Yb

Hence, co-operative luminescence involves emission of a single visible photon due to simultaneous de-excitation of two excited NIR photons of Yb^{3+} ions. The co-operative upconversion photon energy is therefore equal to the double the energy of NIR single photon. Co-operative transitions are generally observed due to either induced by the electronic coulomb interaction among sensitizer (Yb^{3+}) ions on the ground and excited levels, or a single electron operator has a feeble matrix factor for a two-center electronic transition, provided that no overlap among the interrelating ions [43].

In addition to the blue emission from Tm^{3+} ion, Fig. 7a, b also presents green luminescence in the 530–580 nm spectral region, which are not seen in the Stokes spectra (see the inset of Fig. 5). This is due to the existence of different Ln^{3+} ions such as Er^{3+} , Ho^{3+} , etc., that are incorporated in the composition as impurities, in LuSGG:Tm–Yb nanorods. The trace impurity elements were involuntarily commenced through the sample preparation as they are associated with the lanthanide nitrates. To know the concentration of impurities, once again, EDS was taken in view of the impurity ions. From the results, it is observed that the elements are uniformly distributed over the sample and the impurities are found less than 0.5 at %. Similar type of impurity peaks have also been observed in our previous report of LuSGG: Yb^{3+} [26] and in $\text{Pr}^{3+}/\text{Yb}^{3+}$ -doped lanthanide tetraphosphates [41]. The excited state of Yb^{3+} ion around $10,000\text{ cm}^{-1}$ range is well matched with Er^{3+} and Ho^{3+} ions, and hence, resonant and/or non-resonant energy transfer takes place from Yb^{3+} ion to these Ln^{3+} ions [44, 45]. Therefore, the de-excitation peaks observed at 540 and 554 nm due to ($^2\text{H}_{11/2}, ^4\text{S}_{3/2}$) $\rightarrow$ $^4\text{I}_{15/2}$ transitions of Er^{3+} ions and 566 nm belongs to $^5\text{S}_2 \rightarrow ^5\text{I}_8$ transition of Ho^{3+} ion. These luminescence transitions are caused by the efficient resonant energy transfer (ET) from $\text{Yb}^{3+}/\text{Tm}^{3+}$ to $\text{Er}^{3+}/\text{Ho}^{3+}$. When excited by NIR light, two- or three-step energy transfer takes place from Yb^{3+} ions to gives raise to visible upconverted luminescence of these ions. As can be seen from Fig. 7b, the intensities of blue and green emission are nearly the same. Therefore, in the present LuSGG host, interaction of sensitizers (Yb^{3+}) ions with $\text{Er}^{3+}/\text{Ho}^{3+}$ ions competes the $\text{Yb}^{3+} \rightarrow \text{Tm}^{3+}$ and $\text{Yb}^{3+} \rightarrow \text{Yb}^{3+}$ energy transfers. In other words, the probability of co-operative luminescence, ETU and ET are nearly same in LuSGG:Tm–Yb garnet nanorods.

It is also noticed that from Fig. 7a, b, the overall intensity of blue emission is increased up to 5 mol% of Yb^{3+} ion concentration and then decreased, whereas with pump power,



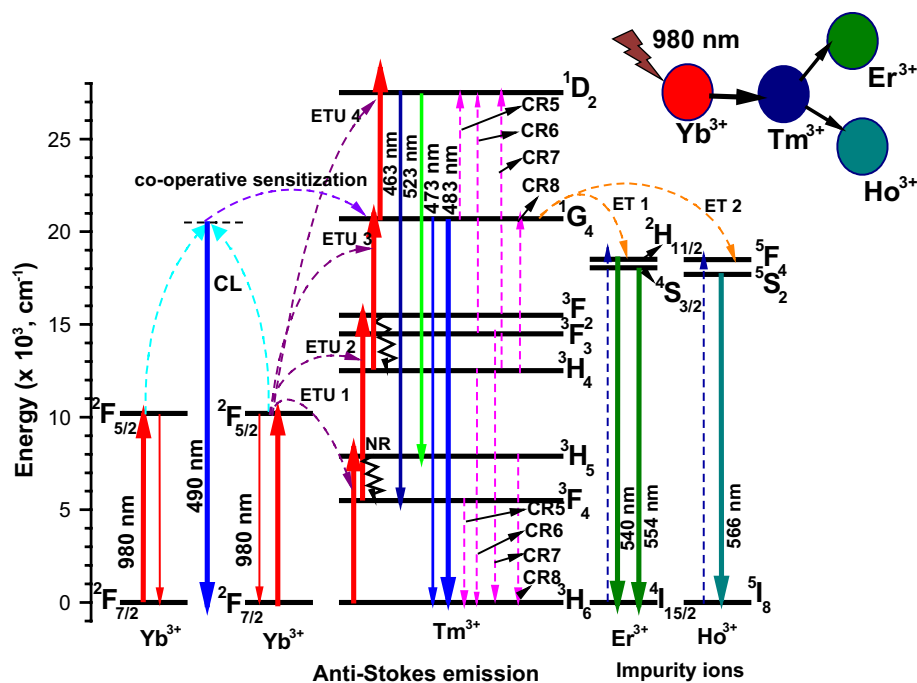
it is continuously increased. On the other hand, the intensity of green emission is decreased with Yb^{3+} ion concentration and increased with excitation power, which is also an evidence for the energy transfer from Yb^{3+} ion $\rightarrow \text{Er}^{3+}/\text{Ho}^{3+}$ ions. From the upconverted luminescence intensities (not shown), it is observed that intensity variation with pump power is different for different peaks. It is followed the order $490 > 540 + 554 > 473 + 483 > 566 > 523 > 463$ nm. With pump power, the intensities of 463 and 523 nm are slightly changed, whereas significant change is observed in the intensity of 490 nm. This indicates that with exciting powers, energy transfer between Yb^{3+} ions, associated with co-operative luminescence, is more stronger/probable than that between Yb^{3+} and Tm^{3+} ions in the LuSGG:Tm-Yb garnet nanorods. Similarly, with respect to Yb^{3+} ion concentration, drastic changes in the peak intensities of 490, 473 and 483 nm are noticed, whereas marginal changes are observed in the other peak intensities. The intensity of co-operative luminescence mainly depends on the distance between the Yb^{3+} ions and is therefore correspond to the degree of Yb^{3+} ion clusters and concentration [46]. It is known that co-operative luminescence is a very good signature of formation of Yb^{3+} ion pairs at distances of less than 5 Å [47]. The increase in upconversion luminescence intensity with Yb^{3+} ion concentration and pump power is due to the enhancement of energy transfer processes from Yb^{3+} ion $\rightarrow \text{Tm}^{3+}/\text{Yb}^{3+}$ ion. At higher concentrations of Yb^{3+} ions, concentration quenching through energy migration among the Yb^{3+} ions and then to traps could lead to the decrease in UC luminescence intensity. Even though the energy transfer upconversion to impurity ions is well known to be 10^5 times more efficient than co-operative luminescence for the same sensitizer (Yb^{3+}) $\rightarrow$ sensitizers (Yb^{3+}) distances [48], as observed in our previous report of LuSGG: Yb^{3+} [26], addition of Tm^{3+} ions into the LuSGG:Yb reduced the ET ($\text{Yb} \rightarrow \text{Er}/\text{Ho}$) and enhanced the ETU ($\text{Yb} \rightarrow \text{Tm}$) and co-operative luminescence ($\text{Yb} \rightarrow \text{Yb}$).

It is clear that the PL intensity is proportional to n th power of excitation laser power for an unsaturated upconversion process which can be represented as $I \propto P^n$, where I corresponds to the intensity of the fluorescence, P represents pump power and the integer n stands for number of absorbed NIR photons per upconverted visible photons [49]. Figure 7d demonstrates the sequence of integrated luminescence intensity versus laser excitation pump power. The slopes (n value) for the 463, 473 + 483, 490, 523, 540 + 554 and 566 nm emissions are found to be 2.8, 2.8, 2.8, 3.6, 2.8 and 3.6, respectively, which are attained by well-known Gaussian fitting (see Fig. 7c). This indicates that four- and three-excited photons (Yb^{3+} - Yb^{3+} ion pairs) are involved for the emissions from $^1\text{D}_2$ and $^1\text{G}_4$ levels of Tm^{3+} ion [50]. Though the slope for the 490 nm exceeds 2, it is understood that two photons are involved in the co-operative luminescence.

Slightly higher value in slope could be due to the overlap of 490 nm peak with the other blue emission peaks. This may also be the reason in observing the variation in the slope values for both 463 and 523 nm, though they are originating from the same $^1\text{D}_2$ level.

Figure 8 depicts the energy-level diagram of anti-Stokes luminescence of LuSGG:Tm-Yb garnet nanorods along with the possible upconversion mechanisms. In the case of upconversion, the Yb^{3+} ions are initially get excited to $^2\text{F}_{5/2}$ level with the 980 nm excitation wavelength as the absorption cross-section of Yb^{3+} ions at this wavelength is very high compared to Tm^{3+} ions. Then, the Yb^{3+} ions transfer their energy to the nearby ground-state Tm^{3+} ions via sequential energy transfer processes. After absorbing this energy, Tm^{3+} ions are promoted to the $^3\text{H}_5$ level followed by de-excitation to the $^3\text{F}_4$ level through non-radiative relaxation. Subsequently, from the $^3\text{F}_4$ level, these ions are further migrated to $^3\text{F}_2$ level by absorbing one more 980 nm photon from Yb^{3+} . The Tm^{3+} ions in the $^3\text{F}_2$ level are relaxed to lower lying $^3\text{H}_4$ level non-radiatively, and then, they are promoted to upper $^1\text{G}_4$ level by absorbing 980 nm photon. Finally, the population of $^1\text{G}_4$ can also be further promoted to $^1\text{D}_2$ level by taking fourth photon at 980 nm. In Fig. 8, these energy transfers are denoted by ETU 1 ($^3\text{H}_6 \rightarrow ^3\text{H}_5$), ETU 2 ($^3\text{F}_4 \rightarrow ^3\text{F}_2$), ETU 3 ($^3\text{H}_4 \rightarrow ^1\text{G}_4$) and ETU 4 ($^1\text{G}_4 \rightarrow ^1\text{D}_2$) transitions followed by non-radiative transitions to the lower levels denoted by NR. Though the observed slope for the 523 nm is near to 4, due to the energy mismatch between the $^2\text{F}_{5/2} \rightarrow ^2\text{F}_{7/2}$ of Yb^{3+} and $^1\text{G}_4 \rightarrow ^1\text{D}_2$ of Tm^{3+} transitions, the transition probability of a fourth Yb^{3+} excitation photon is very low. Therefore, it is proposed that $^1\text{D}_2$ level of Tm^{3+} ion gets populated by the cross-relaxation channels **CR5**: ($^3\text{F}_4, ^1\text{G}_4 \rightarrow ^3\text{H}_6, ^1\text{D}_2$), **CR6**: ($^3\text{H}_4, ^3\text{F}_3 \rightarrow ^3\text{H}_6, ^1\text{D}_2$) and **CR7**: ($^3\text{F}_3, ^3\text{H}_4 \rightarrow ^3\text{H}_6, ^1\text{D}_2$), which are responsible for the emissions at 463 and 523 nm in the upconversion spectra. On the other hand, in addition to this multi-step ETU, two photons can also excite the Tm^{3+} ions to the $^1\text{G}_4$ level via co-operative sensitization [51]. Here, the energy of the two excited Yb^{3+} ions is transferred non-radiatively and quasi-resonantly to a single Tm^{3+} ion. Both mechanisms (multi-step ETU and co-operative sensitization) can also co-exist simultaneously in $\text{Yb}^{3+}/\text{Tm}^{3+}$ co-doped materials and their probabilities depend on the concentration of the dopants and excitation density. As the slope value for 483 nm peak is around 3, hence it is confirmed that multi-step ETU is more dominant than co-operative sensitization in LuSGG:Tm-Yb nanorods. In addition to this, the $^1\text{G}_4$ level can also be populated by the cross-relaxation channel **CR8**: $^3\text{H}_5, ^3\text{H}_4 \rightarrow ^3\text{H}_6, ^1\text{G}_4$. Hence, from the Stark splitting of $^1\text{G}_4$ level, radiative transition $^1\text{G}_4 \rightarrow ^3\text{H}_6$ takes place that results two emissions at 473 and 483 nm in the blue region. Furthermore, two excited Yb^{3+} ions are co-operatively coupled together (denoted by CL) and reach a virtual state followed by the emission at

Fig. 8 Partial energy-level diagram showing anti-Stokes emission and upconversion mechanisms in LuSGG:Tm–Yb nano-garnet rods. Solid lines show the excitation and emission transitions, dashed lines represent the cross-relaxation (CR) (pink), energy transfer upconversion (ETU) processes (purple) and energy transfer (ET) processes (orange), co-operative sensitization (violet) and co-operative luminescence processes (CL) (cyan) and impurity ion excitation (navy blue) and zig-zag lines non-radiative (NR) processes (black)



490 nm. Furthermore, energy transfer from Yb^{3+} to impurities like $\text{Er}^{3+}/\text{Ho}^{3+}$ (denoted by ET) leads to the emissions at 540, 554 and 566 nm. From the slopes, it is confirmed that these processes also involve three photons. However, in our earlier report of LuSGG: Yb^{3+} [26], only two photons were involved; therefore, it is concluded that in the present case,

Table 1 CIE colour co-ordinates (x , y) and correlated colour temperatures (CCT, °K) for the Stokes and anti-Stokes spectra

S. no.	Sample	CIE co-ordinates		CCT (°K)
		x	y	

Stokes ($\lambda_{\text{exc}}=360\text{ nm}$)				
1	LuSGG:1Tm	0.153	0.036	1635
2	LuSGG:1Tm,1Yb	0.139	0.054	1676
3	LuSGG:1Tm,5Yb	0.145	0.058	1676
4	LuSGG:1Tm,10Yb	0.146	0.062	1702
Anti-Stokes				
Concentration-dependent ($\lambda_{\text{exc}}=980\text{ nm}$) $P=1130\text{ mW}$				
5	LuSGG:1Tm,1Yb	0.261	0.460	7516
6	LuSGG:1Tm,5Yb	0.268	0.415	7689
7	LuSGG:1Tm,10Yb	0.194	0.328	15,071
Power-dependent LuSGG:1Tm,1Yb ($\lambda_{\text{exc}}=980\text{ nm}$)				
Excitation power (mW)				
8	398	0.321	0.480	5778
9	544	0.289	0.460	6677
10	676	0.277	0.460	7094
11	798	0.269	0.456	7355
12	914	0.264	0.455	7439
13	1020	0.262	0.457	7507
14	1130	0.261	0.461	7507

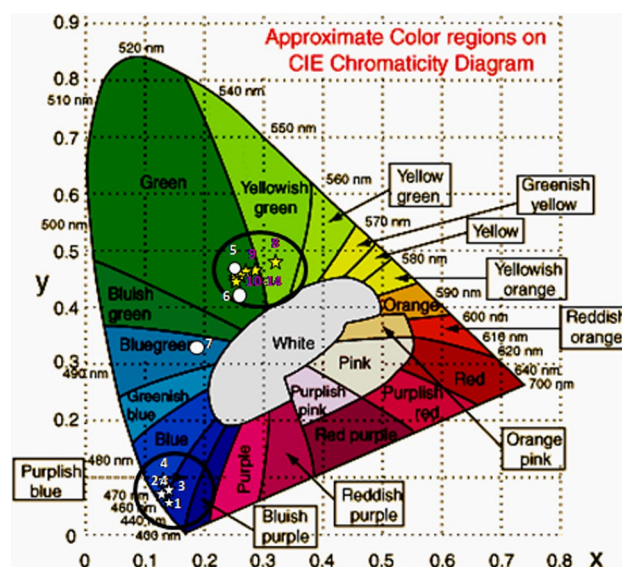


Fig. 9 CIE diagram of Tm–Yb co-doped LuSGG nanorods. Colour co-ordinates of the luminescence spectra excited with **a** 360 nm, represented by white stars (1–4), **b** 980 nm, for various concentrations, represented by white circles (5–7), and **c** 980 nm, for various powers, represented by yellow stars (8–14) moved towards dark green side with increasing power

energy is being transferred from $\text{Yb}^{3+} \rightarrow \text{Tm}^{3+} \rightarrow \text{Er}^{3+}/\text{Ho}^{3+}$ ions. This energy transfers are represented by ET 1 and 2 in Fig. 8. At low Yb^{3+} ion concentration and low excitation regime, ETU dominates. Once the excitation intensity rises, co-operative sensitization and luminescence rises, as well.

Table 2 Comparison of colour co-ordinate (x , y) values in various Tm^{3+} , Yb^{3+} -doped phosphors

S. no.	Name of the compound	Colour co-ordinates		References
		x	y	
1	LuSGG:1Tm ³⁺ ,1Yb ³⁺	0.321	0.480	Present work
2	Zirconium titanate:Tm ³⁺ ,Yb ³⁺	0.218	0.184	[32]
3	MgWO ₄ : Tm ³⁺ ,Yb ³⁺	0.10	0.11	[56]
4	CaZrO ₃ : Tm ³⁺ ,Yb ³⁺	0.19	0.18	[57]
5	BaLa ₂ ZnO ₅ :0.75Tm ³⁺ ,10Yb ³⁺	0.154	0.173	[33]
6	LaGaO ₃ : Tm ³⁺ ,Yb ³⁺	0.088	0.158	[58]
7	GdOCl: 0.1Tm ³⁺ ,10Yb ³⁺	0.143	0.033	[59]
8	NaGdF ₄ :Tm ³⁺ , Yb ³⁺ , Li ⁺	0.29	0.31	[60]

3.3.3 Chromaticity colour

Chromaticity colour co-ordinates on the Commission International de l'Eclairage (CIE) diagram for LuSGG:Tm–Yb garnet nanorods estimated from the photoluminescence spectra are shown in Fig. 9, and the results are also shown in Table 1. These colour co-ordinates are particularly helpful in identifying the luminescence colours. The samples excited by 360 nm are purely exhibiting blue colour (1–4), whereas the samples excited by 980 nm cover green and blue–green region (5–7) of CIE diagram. With varying pump power, the phosphors showed yellow green and green colour (8–14). It is observed from Fig. 9 that, changing the excitation wavelength, the colour co-ordinates are drastically changed compared to the change with respect to concentration and pump power. The points 6, 7 and 8 are closure to the white light region. These results also suggest that addition of small amount of other Ln^{3+} ions like Eu^{3+} or Sm^{3+} will shift the points into the white light region which can be useful for the white light generation and it would be our future work. In addition, with the help of CIE co-ordinates, colour correlated temperature (CCT) values of the LuSGG:Tm–Yb nanorods excited by both 360 and 980 nm were calculated by McCamy empirical formula [52].

The CCT values of LuSGG:Tm–Yb garnet nanorods are given in Table 1. In general, warm white light with CCT around 5000 K is well liked in solid-state lighting applications [53]. Table 1 presents a very eye-catching warm white light with CCT value of 5778 K, achieved by pumping the LuSGG:1Tm–1Yb garnet nanorods at 398 nm (point 8) which is closure to that of the day light (5500 K). The CCT values around 10,000 K may be useful in car head lights or for aquariums with saltwater as the high temperature supports coral reef growth [54]. Emission at 460 nm could be useful for actinic lighting that promotes the photochemical effects including photosynthesis and wastewater treatment [55]. Table 2 compares the colour co-ordinate values with

the other reported $\text{Tm}^{3+}/\text{Yb}^{3+}$ -doped hosts [32, 33, 56–60]. Furthermore, these CIE colour co-ordinates and CCT values of LuSGG:Tm–Yb garnet nanorods can also be additionally optimized by doubly doped some other Ln^{3+} ions such as $\text{Eu}^{3+}/\text{Sm}^{3+}/\text{Er}^{3+}$ ions.

4 Conclusions

LuSGG:Tm³⁺–Yb³⁺ co-doped garnet nanorods have been synthesized by sol–gel method and characterized by X-ray diffraction, scanning electron microscope, energy-dispersive spectra, diffuse reflectance and photoluminescence measurements. The prepared samples were well crystallized in a cubic garnet structure with a crystallite size of 30 nm, and Tm^{3+} and Yb^{3+} ions were occupied the Lu^{3+} dodecahedral sites without distorting the host structure. The lattice parameter of LuSGG:Tm–Yb was found slightly higher than the LuGG:Tm–Yb. SEM images confirmed the formation of nanorods with an average length of 8 μm and diameter of 90 nm. The partial energy-level diagram of Tm^{3+} ions in LuSGG nanorods was established in the range 300–2000 nm. Under 360 nm excitation wavelength, the Stokes luminescence spectra contained strong peaks at around 454 nm corresponding to the $^1\text{D}_2 \rightarrow ^3\text{F}_4$ transition. The decay curves for the $^1\text{D}_2$ level exhibited non-exponential nature and the lifetime was found to be around 20 μs . Visible anti-Stokes luminescence spectra were recorded by exciting the samples with 980 nm. The spectra consisted of two bands in the blue and green region with several peaks. Blue emission peaks are attributed to the energy transfer from Yb^{3+} ion $\rightarrow \text{Tm}^{3+}/\text{Yb}^{3+}$ ions, whereas green emission (530 nm onwards) corresponds to energy transfer from $\text{Yb}^{3+} \rightarrow \text{Tm}^{3+} \rightarrow \text{Er}^{3+}/\text{Ho}^{3+}$ impurity ions. Strong co-operative luminescence at 490 nm was observed under the 980 nm excitation. Optimum concentration was found to be 5 mol% in the LuSGG:Tm–Yb nanorods. Three- photons were involved in populating the $^1\text{G}_4$ and $^1\text{D}_2$ levels in addition to the cross-relaxation channels present in the Tm^{3+} ion. Addition of Tm^{3+} ions in the LuSGG:Yb promotes the ETU and co-operative luminescence processes between $\text{Yb}^{3+} \rightarrow \text{Tm}^{3+}/\text{Yb}^{3+}$ ions and reduces the ET processes between $\text{Yb}^{3+} \rightarrow \text{Er}^{3+}/\text{Ho}^{3+}$ impurities. At low excitation regime and low Yb^{3+} ion concentration, ETU dominates. As soon as the excitation intensity goes up, co-operative sensitization and luminescence rises as well. Colour co-ordinates and CCT values indicated that LuSGG:1Tm–1Yb garnet nanorods at 398 mW power are best suitable for the solid-state lighting applications.

Acknowledgements Dr. R. Praveena is grateful to the DAE-BRNS, Govt. of India for the award of Major Research Project (no. 2012/20/34/11/BRNS/2997). Dr. V. Venkatramu is grateful to DST, New Delhi for the sanction of research project (no. INT/PORTUGAL/P-04/2017) under India–Portugal Bilateral Scientific

and Technological Cooperation. Dr. V. Lavín thanks Ministerio de Ciencia e Innovación (MICIIN) and EU-FEDER for the Project under the National Program of Sciences and Technological Materials (PID2019-106383GB-C44). The authors are also thankful to Sri Venkateswara University (Tirupati, India) for providing characterization facilities.

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